

AI in Culture and Arts – Tech Crash Course

Module Introduction

Benedikt Zönnchen
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MUC.DAI
Munich Center for
Digital Sciences and AI



myt Hochschule
für Musik und Theater
München

1. What is AICA?

2. Course Content

3. Course Schedule

4. Evaluation

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What is AICA?



AI in Culture and Art
BIDT Digitalisierungskolleg



Bayerisches Forschungsinstitut
für Digitale Transformation

EIN INSTITUT DER BAYERISCHEN AKADEMIE DER WISSENSCHAFTEN

- Digitalization college
- Inclusive approach to digitalization
- Interdisciplinarity
- Project-based approach

aica-wavelab.github.io

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Course Content: Three Blocks (provisional plan)

Block 1: Introduction to interactive machine learning:

- (1) How to model intelligence? (2) When is learning possible? (3) How do machines learn? (4) How do we train machines?
- Understanding by experiencing & by (mostly executing) code
- Build your first neural network with Python

Block 2: Generating new things:

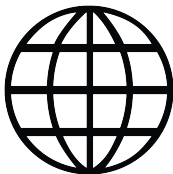
- A probabilistic understanding of machine learning and the free energy principle
- Variational autoencoders (VAEs), diffusion models, recurrent neural networks, transformers
- Generating text, images, audio and more

Block 3: Using large models:

- How neural networks compute semantically rich representations
- Using existing models: fine-tuning, transfer learning (e. g. LoRA), prompt tuning, prefix tuning, knowledge distillation, RAG

I will adapt the pace and content depending on my experience of each bloc!

If you have any interests or topics you want to go into, please feel free to ask me about it. The content is not fixed. There is some flexibility.



Course website

News and announcements

All information of what you have to do!

Relevant links

<https://aica-wavelab.github.io/tech-crash-course/>



GitHub repo

Source code

Tutorial material

Homework material

<https://github.com/aica-wavelab/aica-assignments>

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- **Block 1:** 22nd, 23rd April
- **Block 2:** 20th, 21th May
- **Block 3:** 17th, 18th June
- **Submission Deadline:** 10th July

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The most important criteria for me is that you try!

- **Attendance:** Minimum 4/6 days (first bloc is probably the most important one).
- **In class tutorial:** You must follow the instructions of the tutorial (link) and hand-in what is stated there. You are incentivised to work together.
- **Homework:** Some of the tutorials are intended as homework. In general, you should finish the tutorial at home if needed.

Depending on your institution, your homework will differ. HMTM students can always do the HM homework instead.

Submit your solutions **here!**

Reminder to myself: Write down the password.

Any questions?